

केन्द्रीय विद्यालय संगठन, कोलकाता संभाग

KENDRIYA VIDYALAYA SANGATHAN, KOLKATA REGION

प्री-बोर्ड परीक्षा / PRE-BOARD EXAMINATION – 2025-26

कक्षा/CLASS – X

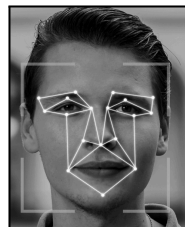
विषय/SUB. – Artificial Intelligence (417)

अधिकतम अंक/MAX. MARKS – 50

समय/TIME – 02 घंटे/Hours

SECTION A: OBJECTIVE TYPE QUESTIONS

1	Answer any 4 out of the given 6 questions on Employability Skills(4 x 1 = 4)	
i.	Which of these is NOT a common communication barrier? a) Linguistic barrier b) Interpersonal barrier c) Financial barrier d) Organizational barrier	1
ii.	The “SMART” method can be used to set goals to make you successful in your career and personal life. What does 'M' in “SMART” stand for? a) Motivator b) Measurable c) Maintainable d) Mentor	1
iii.	Pranjali gets up at 5 am and goes to her badminton classes. Then she comes home and finishes her homework before going to school. She does this all by herself. No one tells her to do it. This is an example of a) Self-motivation b) External c) Both (a) and (b) d) Not any specific type of motivation	1
iv.	Software that starts working as soon as we switch on a computer is a.) RAM. b.) ROM c.) Operating System d.) None of the above	1
v.	Which of the following are misconceptions about entrepreneurs? a) Entrepreneurs are born, not made. b) A person having a big business is an entrepreneur. c) A person needs a lot of money to start a business. d) All of the above	1
vi.	Choose the option which defines sustainable development. a) Taking care of future generations. b) Taking care of only ourselves. c) Taking care of ourselves and the future generations. d) None of the above	1
2	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	Results based on discriminatory assumptions made while developing algorithms by the developer are called _____. a) AI Ethics b) AI Bias c) Test Data d) Training Data	1
ii.	Assertion: Ethics provide guidance in distinguishing right from wrong. Reasoning: Ethics consist of a set of values and morals that aid individuals in making moral judgments and decisions. a) Both Assertion and Reasoning are true, and Reasoning is the correct explanation of the Assertion. b) Assertion is true, but Reasoning is false. c) Both Assertion and Reasoning are true, but Reasoning is not the correct explanation of the Assertion. d) Assertion is false, but Reasoning is true.	1
iii.	In which type of machine learning is the data labeled with the desired output? a) Supervised Learning b) Unsupervised Learning c) Reinforcement Learning d) Deep Learning	1
iv.	_____ is an integral part of the model development process that helps to find the best model that represents our data and how well the chosen model will work in the future.	1
v.	_____ is mainly focused on enhancing raw input images or preparing them for other tasks, such as rescaling, correcting brightness, or changing tones. It is a subset of _____. a) Computer Vision; Image Processing b) Machine Learning; Artificial Intelligence c) Image Processing; Computer Vision d) Object Detection; Image Processing	1
vi.	If an image has resolution 1280x1024, it means:	1

	a.) The file size is 1280 KB c.) Each pixel is 1280 bytes large	b.) 1280 pixels across width and 1024 pixels in height d.) The image is grayscale only																
3.	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)																	
i.	The method of collecting correct and dependable data to work with is known as _____. a) Problem Scoping b) Data Acquisition c) Data Exploration d) Modeling		1															
ii.	Consider the following data where we are trying to predict what fruit it is based on the color of the fruit. <table border="1"><thead><tr><th>Fruit</th><th>Color</th><th>Price</th></tr></thead><tbody><tr><td>Apple</td><td>Red</td><td>\$1.8</td></tr><tr><td>Orange</td><td>Orange</td><td>\$2</td></tr><tr><td>Banana</td><td>Yellow</td><td>\$1</td></tr><tr><td>Grape</td><td>Purple</td><td>\$3</td></tr></tbody></table> The column “Fruit” is the a) Instance b) Label c) Model d) Feature		Fruit	Color	Price	Apple	Red	\$1.8	Orange	Orange	\$2	Banana	Yellow	\$1	Grape	Purple	\$3	1
Fruit	Color	Price																
Apple	Red	\$1.8																
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iii.	In a medical test for a rare disease, out of 1000 people tested, 50 actually have the disease while 950 do not. The test correctly identifies 40 out of the 50 people with the disease as positive, but it also wrongly identifies 30 of the healthy individuals as positive. What is the accuracy of the test? a) 90% b) 92% c) 96% d) 98%		1															
iv.	Assertion: The sum of the values in a confusion matrix's row represents the total number of instances for a given actual class. Reasoning: This enables the calculation of class-specific metrics such as precision and recall, which are essential for evaluating a model's performance across different classes. Choose the correct option: (a) Both A and R are true and R is the correct explanation for A (b) Both A and R are true and R is not the correct explanation for A (c) A is True but R is False (d) A is false but R is True		1															
v.	Identify the application of Computer Vision from the given picture: a) Facial Recognition b) CV in Retail c) Medical Imaging d) Face Filters		1															
vi.	Count and write the number of tokens in the sentence: “artificial intelligence is a technique that enables computers to mimic human intelligence.”		1															
4.	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)																	
i.	What do frameworks provide in the context of problem-solving? a) Step-by-step guidance b) Random solutions c) Legal advice d) Ethical justifications		1															
ii.	Assertion (A): Information processing in a neural network relies on weights and biases assigned to nodes. Reasoning (R): These weights and biases determine how strongly a node is influenced by its inputs and its overall contribution to the next layer. Which of the following is correct? a) Both A & R are correct. b) Both A & R are incorrect. c) Only A is correct. d) Only R is correct.		1															
iii.	When we use the training data for testing the model, the model is said to be a) Overfitting. b) Under fitting. c) Perfectly fit. d) None of these		1															
iv.	Syntax refers to the grammatical structure of a sentence. (True/False)		1															

v.	Consider the below given example of a Perceptron, a single layer ANN Model. The model considers four key attributes, each assigned a weight based on importance. Perceptron Example (go to the park) 1 Do I have a jacket? 1.5 0 Do I have an umbrella? 1.0 1 Is it sunny now? 3.0 0 What is the weather forecast for later? 2.5 4.0 Y B  YES  NO  YES  NO A bias(b) value of 4.0 is added. The formula used is: $y = (w_1 \times x_1) + (w_2 \times x_2) + (w_3 \times x_3) + (w_4 \times x_4) + (w_b \times b)$ What will be the value of y for the given scenario? a) 0.4 b) 1.4 c) 0.5 d) 0.9	1										
vi.	The term that refers to the process of detecting instances of the objects, giving them a category, and then giving each pixel a label based on that is . a) Image Classification b) Instance Segmentation c) Object Detection d) Feature Extraction 	1										
5. Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)												
i.	Which of the following statements is NOT true about supervised learning? a) Requires labeled data for training. b) Used for classification and regression tasks. c) Can be less efficient for large datasets. d) Often used in image recognition applications.	1										
ii.	Consider the sample test results given for an image classification model below: <table><tr><th>Prediction</th><th>Reality</th></tr><tr><td>Cat</td><td>Dog</td></tr><tr><td>Dog</td><td>Cat</td></tr><tr><td>Dog</td><td>Dog</td></tr><tr><td>Dog</td><td>Cat</td></tr></table> What is the error rate of this prediction? (rounded to two decimal places) a) 0.57 b) 0.89 c) 0.25 d) 0.75	Prediction	Reality	Cat	Dog	Dog	Cat	Dog	Dog	Dog	Cat	1
Prediction	Reality											
Cat	Dog											
Dog	Cat											
Dog	Dog											
Dog	Cat											
iii.	In the byte image format, what is the range of possible pixel values? a) 0 to 10 b) 0 to 100 c) 0 to 1000 d) 0 to 255	1										
iv.	_____ is the outcome of the model rightly predicting the negative class as negative class. a) False Positive b) True Positive c) False Negative d) True Negative	1										

v.	Which type of bot is complex to create, works based on any instructions, requires good language processing skills, and has wide functionality? a) Script Bot b) Smart Bot c) Cleaning Bot d) Talbot	1
vi.	In text pre-processing a word “flies” is converted to “fli”. This is done in _____ (lemmatization/stemming)	1

SECTION B: SUBJECTIVE TYPE QUESTIONS

Answer any 3 out of the given 5 questions on Employability Skills (2 x 3 = 6 marks)

Answer each question in 20 – 30 words.

6.	Mention any two positive facial expressions which you can use in making effective communication.	2
7.	What are the different types of stress management techniques?	2
8.	What is the difference between a file and a folder?	2
9.	How does an entrepreneur bring changes in society?	2
10.	How is organic farming contributing towards sustainable development?	2

Answer any 4 out of the given 6 questions in 20 – 30 words each (2 x 4 = 8 marks)

11.	Write the principles of Bioethics.	2
12.	Identify the model: Classification or Regression? Case 1: Predicting whether a customer is eligible for a bank loan or not? Case 2: Predicting weather for next 24 hours	2
13.	Differentiate between Classification and Clustering.	2
14.	Understanding both error and accuracy is crucial for effectively evaluating and improving AI models.”. Draw the confusion matrix for the following data: The number of True positive = 100 The number of True Negative = 47 The number of False Positive = 62 The number of False Negative = 290	2
15.	What is a "byte image" format in the context of digital images?	2
16.	What will be the results of conversion of the term, ‘happily’ and ‘caring’ in the process of stemming and lemmatization?	2

Answer any 3 out of the given 5 questions in 50– 80 words each (4 x 3 = 12 marks)

17.	Read the case study below and answer the following questions: A school develops an AI system to shortlist students for a competitive scholarship. The algorithm is trained to prioritize students who complete online application forms quickly — assuming that faster completion reflects confidence, competence, and tech- savviness. However, students who are not fluent in English, or who do not have access to a computer at home, take longer to fill out the form. As a result, many deserving students are unfairly rejected by the algorithm. A. Identify two reasons why the algorithm gave biased results. B. Mention two bioethics principles that can help solve such a problem and explain how they apply.	4
18.	A healthcare provider wants to improve patient care by predicting the length of hospital stays for different medical conditions. They have a dataset containing patient demographics, medical history, and treatment details. The task involves: A. To predict whether a patient will have a short or long hospital stay. B. To predict the number of days a patient will stay in the hospital.	4

	C. To segment patients into groups with similar characteristics for personalized treatment plans. D. To identify patterns in patient treatments and outcomes. Identify the type of model (classification, regression, clustering, and association model) in the above tasks.	
19.	Read the following paragraph and answer the questions that follow: A school recently tested an AI model designed to predict whether students would pass or fail their final exams. Out of 100 students, the model correctly predicted that 40 students would pass and they actually did. It also correctly identified 30 students who were going to fail. However, the model predicted that 20 students would pass, but they ended up failing. Additionally, it predicted that 10 students would fail, but they actually passed. A. Draw the confusion matrix based on the above information. (2) B. Calculate the accuracy of this classification model. Show your working. (1) C. Write the total number of wrong predictions made by the model. (1)	4
20.	Identify the name of the application of Machine Learning (ML) or Deep Learning (DL) being used in the following scenarios: A. A banking app monitors your usual transaction habits. One day, it detects a sudden large withdrawal from a foreign country and instantly alerts you that this might be suspicious activity. B. A traffic monitoring system uses AI to analyze live road footage and categorize vehicles into types such as “car,” “bus,” “bike,” and “truck.” C. A retail store uses smart cameras that scan items on shelves and automatically display their names — such as “Bread,” “Juice Bottle,” and “Chocolate Bar.” D. A tablet-based exam app allows students to write numbers on the screen using a stylus, and the system recognizes and converts their handwritten numbers into digital text automatically.	4
21.	Read the following three documents and answer the question that follow: Document 1: "Students love studying AI" Document 2: "AI is transforming education" Document 3: "Teachers and students explore AI tools" Create the document vector table for the above documents applying the “Bag-Of-Words” Algorithm of NLP.	4

*****BEST OF LUCK*****